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What Do We Know about Geographical Knowledge Spillovers and Regional Growth?: A Survey of the Literature

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DÖRING T. and SCHNELLENBACH J. (2006) What do we know about geographical knowledge spillovers and regional growth?: a survey of the literature, *Regional Studies* **40**, 375–395. Modern (endogenous) growth theory tells us that knowledge is crucial for the sustained growth of high-income economies. Against this background, the paper provides a survey of theoretical and empirical findings highlighting the question of how geographically limited knowledge diffusion can help to explain clusters of regions with persistently different levels of growth. It discusses this topic in two steps. First, the theoretical concept of knowledge spillovers is outlined by discussing the different types of knowledge, the spatial dimension of knowledge spillovers, and the geographical mechanisms and structural conditions of knowledge diffusion. Second, it analyses the empirical evidence concerning the theoretical propositions.

Knowledge spillovers Regional growth Research and development Innovation Diffusion of innovations

DÖRING T. et SCHNELLENBACH J. (2006) Qu'est-ce qu'on sait des retombées de connaissance géographiques et de la croissance régionale?: Une revue de la documentation, *Regional Studies* **40**, 375–395. La théorie de croissance moderne (endogène) indique que la connaissance est essentielle pour la croissance durable des économies à revenus élevés. Sur ce fond, l'article cherche à fournir une revue des conclusions théoriques et empiriques qui mettent l'accent sur la question suivante: comment la diffusion de la connaissance, bien délimitée sur le plan géographique, pourrait-elle aider à expliquer les regroupements des régions dont les taux de croissance diffèrent continuellement? L'article cherche à aborder cette question à deux temps. Primo, on esquisse la notion théorique de retombées de connaissance en discutant des types de connaissance différents, de la dimension géographique des retombées de connaissance, et des mécanismes géographiques et des conditions structurelles de la diffusion de la connaissance. Secundo, l'article cherche à analyser les preuves empiriques concernant les propositions théoriques.

Retombées de connaissance Croissance régionale Recherche et développement Innovation Diffusion des innovations

DÖRING T. und SCHNELLENBACH J. (2006) Was ist über Nebeneffekte geographischer Kenntnisse und Regionalwachstum bekannt? Ein Überblick über die Literatur, *Regional Studies* **40**, 375–395. Die moderne (endogene) Wachstumstheorie behauptet, daß Kenntnisse für anhaltendes Wachstum in Wirtschaften mit hohem Einkommen äusserst wichtig sind. Vor diesem Hintergrund liefert dieser Aufsatz eine Übersicht über theoretische und empirische Befunde, welche ein Schlaglicht auf die Frage werfen, wie geographisch begrenzte Verbreitung von Kenntnissen dazu beitragen kann, Cluster von Regionen mit anhaltend unterschiedlichen Wachstumsgraden zu erklären. Er behandelt dies Thema in zwei Schritten: zunächst wird der theoretische Begriff der Nebenwirkungen von Kenntnissen umrissen, indem die verschiedenen Arten von Kenntnissen, die räumliche Dimension der Nebenwirkungen von Kenntnissen, sowie der geographische Mechanismus und strukturelle Bedingungen der Wissensverbreitung diskutiert werden. Sodann analysiert der Aufsatz die empirischen Beweise hinsichtlich der theoretischen Thesen.

Nebene Wirkung von Kenntnissen regionales Wachstum Forschung und Entwicklung Innovation Verbreitung von Innovationen

DÖRING T. y SCHNELLENBACH J. (2006) ¿Qué sabemos de los desbordamientos geográficos de conocimiento y del crecimiento regional? – Estudio de la literatura, *Regional Studies* 40, 375–395. La teoría moderna del crecimiento (endógeno) postula que el conocimiento es fundamental para el crecimiento sostenido de las economías con altos ingresos. Con estos datos, en este ensayo mostramos un estudio de los resultados teóricos y empíricos y planteamos la cuestión de cómo se puede explicar la existencia de aglomeraciones de regiones que presentan continuamente diferentes niveles de crecimiento por la divulgación geográfica limitada de conocimientos. En este documento analizamos este tema desde dos puntos de vista: Primero, destacamos el concepto teórico de los desbordamientos de conocimiento al examinar los diferentes tipos de conocimiento, las dimensiones espaciales de estos desbordamientos de conocimiento y los mecanismos geográficos y las condiciones estructurales de la divulgación del conocimiento. Segundo, también analizamos las pruebas empíricas con respecto a las proposiciones teóricas.

Desbordamientos de conocimiento Crecimiento regional Investigación y desarrollo Innovación Divulgación de innovaciones

JEL classifications: R11, R12, R58

INTRODUCTION

It has become commonplace among political practitioners and commentators as well as increasingly popular among economists to claim that knowledge is quintessential in increasing competitiveness and fostering economic growth.¹ This view is in concurrence with neoclassical growth theory and its standard result is that once a steady-state is reached, growth in per-capita income can only be induced by a growth in the stock of knowledge, which in turn allows the implementation of a more efficient technology of production (ROSENBERG, 1963; ARROW, 1985; MALECKI and VARAIYA, 1986; JAFFE, 1998; SMOLNY, 2000; BARRO and SALA-I-MARTIN, 1995; AGHION and HOWITT, 1998). This has led more recent contributions to the theory of economic growth to endogenize knowledge-generating processes in order to explain sustained growth, in some cases without employing the metaphor of a steady-state.² These latter approaches share some common ground in their emphasis on positive feedback mechanisms that are needed to explain the absence of diminishing returns, but display some heterogeneity regarding the sources of these mechanisms that are suspected in processes such as learning-by-doing, the accumulation of human capital or the supply of public goods such as publicly funded research. This survey is particularly interested in the spatial dimension of such mechanisms. The main interest lies not in internal learning-by-doing effects and their repercussions in single firms, but in knowledge spilling over from one firm to the other, and in the spatial dimension of this problem: is the spatial range of knowledge spillovers limited, and, if so, why? And what are the implications of such limitations for an economic policy that aims to sustain or foster economic growth?

In the particular case of the spatial distribution of knowledge, both static and dynamic aspects play a role: spatial knowledge spillovers are static externalities in the sense that marginal factor productivities in one production facility are increased by positive externalities from another; but knowledge clustered in a particular

region also potentially produces dynamic externalities, which in turn allow for positive returns to scale in the aggregate production function of this region.³ Positive dynamic externalities, accounting for the fact that experience, which may, for example, diffuse between firms through communication between individuals, are negatively correlated with unit costs, providing a rationale for the assumption of constant or even increasing marginal factor productivities in the aggregate production function, and these in turn allow the modelling of sustained economic growth.

While growth theory has always focused on processes unfolding in time, the introduction of knowledge spillovers by the new growth theory therefore implies an interest in processes that are unfolding in time and space. Both perspectives on knowledge are complementary: while the new growth theory calls attention to the effects of knowledge-related externalities on growth, the literature on spatial knowledge spillovers confronts us with the fact that despite its public good properties, knowledge does not usually diffuse instantaneously to production facilities around the world. Regional patterns of knowledge diffusion, as well as barriers to the diffusion of knowledge, can therefore feature prominently in explaining the differential growth of production and incomes between regions.

This role of knowledge spillovers and the high dispersion of scattered contributions appear to be a good reason to provide a survey of the literature relevant to the problem. Maybe more importantly, spatial knowledge spillovers are of interest to different subdisciplines of economics, such as macroeconomics, public economics and innovation economics. A survey of the field can therefore also serve the purpose of informing researchers of recent results produced in neighbouring fields. The argument will proceed as follows. The second section provides an overview of theoretical contributions to the economics of knowledge and its diffusion. The third section surveys the related empirical literature. The fourth section concludes.

EXTERNALITIES IN THE PRODUCTION OF KNOWLEDGE AND REGIONAL SPILLOVERS

Some economic perspectives on knowledge

Economists do often speak of knowledge, but they have not found a consensus regarding the exact meaning of it. Put differently, 'knowledge' is a term that is regularly used, but its connotations vary with different authors or even schools of thought. Choosing a definition that uses the term in a broad sense, this paper understands knowledge as comprising all cognitions and abilities that individuals use to solve problems, make decisions and understand incoming information. The terms 'knowledge' and 'information' therefore denote separate things, and knowledge is understood as a tool that can be consciously used by individuals – although it certainly can also be used subconsciously, a fact introduced into the economic context by HAYEK (1945). Hayek was also responsible for noting the existence of dispersed knowledge, a dispersion that can occur between individuals and which is a spatial dimension. Thus, it is important also to recognize the modular characteristic of knowledge. Finally, knowledge is also understood as being dependent on time and context (DOHSE, 2001), i.e. it is not static but continually evolving, and the direction of its evolution depends on boundary conditions such as the institutional framework.

While such an encompassing definition of knowledge lays an emphasis on the heterogeneity of knowledge and its evolution over time, it is usually modelled more statically in traditional neoclassical contributions to growth theory. There, knowledge is simply introduced as an input factor into a given production function. However, in contrast to this simple approach, more recent contributions have also laid emphasis on a circular causation that allows for learning from given processes of production in order to explain an endogenous accumulation of knowledge. A growth in the stock of human capital from learning-by-doing and industrial research and development (R&D) processes is explicitly taken into account (AUDRETSCH and FELDMAN, 1996; AUDRETSCH, 1998). These processes are then thought of as being highly path-dependent due to the cumulative nature of knowledge-generating processes (NELSON, 1982; DOSI, 1988; CANIELS, 2000; DOHSE, 2001).⁴

An early attempt to endogenize the growth of knowledge has already been proposed by ARROW (1962), who assumed technological progress to occur through learning-by-doing effects in the production of capital goods. This can be interpreted as an important first step towards endogenous growth theory, which of course has looked not only at the potential of learning-by-doing to explain non-decreasing marginal factor productivities, but also far beyond, e.g. at endogenous innovation or the growth of a stock of

human capital through education. Learning-by-doing in such approaches does not necessarily go hand in hand with knowledge externalities. If it is assumed that knowledge does not spillover at all, then experience effects are limited to single firms. If, however, learning-by-doing is implied in the specification of aggregate production functions, then it is plausible to think of the economic mechanism behind this specification as a combination of knowledge spillovers between single firms and the positive impact of the scale of economic activity on productivity.

The crucial question is therefore how the spilling over of knowledge from one firm to the other can be approached theoretically. In this respect, path-dependency in acquiring knowledge has become a major issue in more recent contributions and features prominently in the evolutionary economics branch of the literature compared with the literature rooted more in the neoclassical tradition. Recognizing that the abilities of individuals to use knowledge that is subjectively new to them and which may be limited by their cognitive history – it is conceivable that they cannot make any reasonable use of new, although perhaps superior, knowledge simply because they lack necessary complementary knowledge, which they have not accumulated on their learning path – is not easy from a neoclassical point of view.

Beyond such possible incompatibilities on the side of the recipient of (subjectively) new knowledge, difficulties in the diffusion of knowledge may also arise due to the attributes of the observed knowledge itself. A common distinction is made between explicit knowledge that can be communicated and tacit knowledge that is often used unconsciously by an individual and cannot be easily made the subject of verbal communication (POLANYI, 1967).⁵ If secrecy about explicit knowledge cannot be enforced or exclusive rights for its application through patent protection cannot be ensured, it has the technical properties of a public good (ROMER, 1990), which is the case for a large fraction of publicly funded scientific research at universities and similar research institutions. Obviously, the incentives to invest private resources into knowledge accumulation differ greatly depending on such technical and institutional conditions.

Given these distinctions, it also becomes easy to distinguish between knowledge and human capital. Human capital comprehends tacit and explicit knowledge that individuals do indeed use, while knowledge itself is a more encompassing category that comprehends the wealth of information and routines that is principally available to individuals. Here, the distinction of POPPER (1972) between objective and subjective knowledge may also be usefully introduced. Objective knowledge is what is written down and waits to be used, while subjective knowledge (here human capital) denotes the stock of knowledge that does indeed affect subjective decision-making. In this sense, only

knowledge can be continuously accumulated over time, while human capital presupposes a selective learning of knowledge by each new generation of individuals. The important question in the present framework is then if and how spatial spillovers of available knowledge do indeed have an influence on human capital across regional boundaries.

Knowledge in space: approaches to the problem

In endogenous models of economic growth-incorporating knowledge, positive externalities are a common feature of processes of knowledge accumulation (ROMER, 1986; DOSI, 1988; BERNSTEIN and NADIRI, 1989; CARLINO, 1995; AUDRETSCH and FELDMAN, 1996; ANSELIN *et al.*, 1997; CANIELS, 2000; KEILBACH, 2000). The social benefit of generating knowledge is generally considered to be higher than the private benefit of this activity,⁶ which partly follows from the technical property of knowledge as being generally non-excludable: whoever may use a knowledge spillover has generally no incentive to compensate the producer of this externality for his/her beneficial activities. For costly R&D activities, this possibility of increasing one's productivity by free-riding on the research of others implies a propensity to underinvest into knowledge-generating activities (AUDRETSCH, 1998). In addition to this short-term effect, knowledge spillovers also do have a dynamic component because the magnitude of their effects on productivity depends on the amount of knowledge that was already accumulated in the past (HENDERSON *et al.*, 1995; HENDERSON, 1997; DOHSE, 2001). This should be kept in mind as a barrier to the diffusion of knowledge in addition to the spatial distance between the producer of an externality and his/her potential recipient.

In the first pioneering work on the geographical dimension of knowledge with a focus on spatial aspects of innovation diffusion, Hägerstrand examined the dissemination of innovations as a spatial process in his monograph of 1953 (for a translation of the work initially published in Swedish, see HÄGERSTRAND, 1967). The main hypothesis that lies at the core of his inquiry is that geographic patterns of variety in human behaviour must be analysed by looking at the knowledge available to the individual decision-maker (individual, firm, etc.), which in turn is thought to depend on a 'social network' of interpersonal communications through which knowledge diffuses. This very early social network approach was certainly a step ahead of later economic approaches, which often relied on less sophisticated assumptions regarding the diffusion of knowledge. This early network approach therefore takes a midway position: knowledge spills over, but not perfectly.

The ends of the continuum of degrees of the mobility of knowledge – perfect mobility and no mobility at all – can also be found in the economic literature

(RICHARDSON, 1973; CANIELS, 2000). A traditional assumption in the earlier neoclassical literature is to view knowledge as being costlessly available to individuals who, given the assumed absence of learning costs and their propensity to act rationally, have an incentive to use the entire available stock of knowledge. Knowledge is entirely disembodied and can be communicated without friction. The implicit assumption of universally costless communication implies that distance does not play a role. A knowledge spillover diffuses instantaneously throughout the entire economy such that technically it is simply a pure public good. This obviously implies that regional differentials of incomes and their growth rates cannot be explained on the grounds of regional divergences in the stock of knowledge, because any region trailing the other can immediately catch up by imitating the more successful technology (for a survey of the discussion in this issue, see KEILBACH, 2000).

The assumption of disembodied knowledge has also been used in early endogenous growth theory (ARROW, 1962; ROMER, 1986), and it still appears to be a core assumption of the neoclassical new growth theory that disembodied knowledge as an essentially non-rival good has a quantitatively important impact on the macro-level through spillovers (ROMER, 1996; critically LANGLOIS, 2001), although firm-specific knowledge thought to be embodied in particular production processes and therefore not a source of spillovers between firms often appears as a complement to disembodied knowledge in modern endogenous growth models.

At the other end of the continuum are models of cumulative causation (MYRDAL, 1964; KALDOR, 1970; MALECKI and VARAIYA, 1986) that work with the assumption of a world in which knowledge spillovers simply do not exist. Knowledge is modelled as a private good that can be used by a clearly confined group of users so that spatial spillovers are not of any interest. Relative advantages in the generation of knowledge and, closely related, in productivity can persist because knowledge-related catching-up is not accounted for. From this perspective and in complete contradiction to the approaches described above, the differential development of regions can then be almost entirely attributed to differences in the stock of knowledge. These are expected to persist and even to increase, because generating additional new knowledge is assumed to be relatively easier the higher is the stock of knowledge already accumulated.

Both these extreme assumptions, complete knowledge spillovers and no knowledge spillovers at all, may be criticized empirically for their lack of realism. That knowledge spreads in the spatial dimension appears to be obvious, but it is equally obvious that such diffusion processes take time and are often incomplete. Theoretical approaches taking this into account usually assume that knowledge is a regional public good with a limited

spatial range. Two basic types of models working with this assumption can be distinguished (RICHARDSON, 1973; CANIELS, 2000): models of epidemic and of hierarchical diffusion of knowledge. In the first case, every region in the neighbourhood of another region in which new knowledge is generated is asserted a positive probability of being the recipient of a knowledge spillover, i.e. of harbouring individuals or businesses that adopt the newly generated knowledge. The diffusion between regions takes place horizontally, on the level of innovators and potential adopters. Contrary to this class of models, hierarchical models are primarily interested in diffusion processes that take place between centres (such as industrial agglomerations) and spread to peripheral regions only later, if at all. The assumption that the probability of adoption is smaller than unity also accounts, apart from the influence of pure distance, for the above-mentioned fact that complementary knowledge necessary to appropriate spillovers is not necessarily present with each potential recipient. Closely related, the simple intuition underlying the higher probability of adoption in centres is the fact that there already are more R&D activities there compared with peripheral regions. It is then much more likely that the necessary complementary knowledge is to be found in the centre. Empirically, both types of knowledge diffusion can be observed and can even occur simultaneously (CANIELS, 2000).

A more sophisticated class of models compared with the two basic types are the so-called leapfrogging models, which synthesize elements from endogenous growth theory and the New Economic Geography and explain the growth of agglomerations with local learning externalities.⁷ Two mechanisms of technological progress are distinguished in these models: an incremental growth of knowledge resulting from learning-by-doing, and growth rate depending positively on the stock of technological knowledge already available. In addition to this incremental growth, occasional large-scale breakthroughs lead to technological paradigm changes and a depreciation of the former stock of knowledge. Furthermore, productivity within the new paradigm exceeds productivity under the old paradigm only after a (possibly rather lengthy) transition period, so that an early change of paradigm is not rational for regions that are sufficiently experienced in using the old technology and which have reached a high productivity level there. A region that is sufficiently inexperienced under the old paradigm with relatively low productivity and income levels has, on the other hand, an incentive to switch paradigms immediately. BREZIS and KRUGMAN (1993, p. 1) call this a 'natural life cycle of urban rise and decline', which hints at the fact that thus far the relatively poor region now operating under the new technological paradigm will not only catch up, but also will eventually be more productive than the other region because their incremental learning

process now takes place within a relatively superior paradigm.

Mechanisms of the spatial diffusion of knowledge

The above discussion has hinted at the fact that knowledge spillovers are of economic relevance because of their impact on regional income growth. Due to the rather sketchy treatment of knowledge spillovers in the approaches discussed above, however, it appears to be necessary to have a closer look at what could be coined the 'microeconomics of knowledge spillovers', i.e. at the micro-level conditions of the spatial diffusion of knowledge.

A common mechanism for a transfer of knowledge is the mobility of individuals and the trade or transfer of goods, which, in one way or another, carry production-related knowledge with them (e.g. MATUSIK and HILL, 1998), and which, notably, would also allow a spilling over of embodied knowledge. A particularly interesting mechanism of embodied knowledge spilling over is the transfer of human capital (i.e. a skilled worker commanding tacit knowledge) from one firm to another. In an early model of this mechanism, PAKES and NITZAN (1983) show that optimal contracts involving stock options can help the entrepreneur to undermine a spillover of embodied knowledge that is unfavourable to him/her by giving the employees an incentive to stay with the firm.

In a similar vein, FOSFURI *et al.* (2001) show that a multinational enterprise that needs to train local employees to run a local production facility can prevent a technology spillover that would occur if the local employee changed his/her employer by granting a pecuniary externality to that employee. Accordingly, a technological spillover to local firms occurs only if the rival local use of the multinational enterprise's technology is not a fierce competitor, i.e. if the multinational enterprise has no incentive to grant his/her employee a higher wage. FOSFURI and RONDE (2004) also show not only that voluntary contracts may reduce the amount of spillovers of embodied knowledge through mobile employees, but also that trade secret laws, which forbid employees to use knowledge acquired with an old employer at their new firm, reduce spillovers and even tend to reduce the incentive to build clusters in the first place. The theoretical results on optimal contracts are particularly interesting from a normative point of view. They show that spillovers of embodied knowledge through the labour market occur only if they increase overall welfare. If they do not, then a contract can always be found that makes the original employer and his/her employee better off if the employee remains at the old firm.

As the discussion by FOSFURI *et al.* (2001) has already insinuated, foreign direct investments by multinational companies may also be an important channel for knowledge spillovers. MARKUSEN and

VENABLES (1999) build a model where the demand for intermediate products by a foreign direct investor can create a local industry for such intermediate products, which, in turn, may even lead to the establishment of local producers of final goods competing in the market of the multinational company. It is easily conceivable that in such a setting the local producer of intermediate goods serves as a node for flows of knowledge from the multinational company to potential local competitors. Finally, it is also conceivable that nothing else than the immaterial knowledge about modes of production spills over from one region into the other through a licensing of patented technologies, through shared research projects, through scientific publications etc. The spatial range of these types of knowledge transfers differs because the costs of transferring pure knowledge through communication over great distances are considerably lower compared with the transfer of goods, individuals or even physical production facilities.

Following CANIELS (2000), some more detailed propositions on the conditions of knowledge diffusion are possible:

- If the source of new knowledge is in the private sector, it will usually release relevant information more reluctantly compared with a source in the public sector: the latter often has the explicit task of producing knowledge as a public good and it therefore tends to circulate the results of its research activities voluntarily.
- The recipient of new knowledge needs to be endowed with the capacities necessary to use available knowledge (COHEN and LEVINTHAL, 1990), which includes his/her cognitive capacities, but also his/her willingness to incur costs of learning new knowledge. For example, the latter can not be presupposed if the recipient is a rationally ignorant voter deciding on how to produce a public good (SCHNELLENBACH, 2004). In other words, it must appear to be economically rational for the recipient to use their new knowledge (COHEN and LEVINTHAL, 1989). If one argues within a model involving not maximizers but rational satisficers, then dissatisfaction with the results produced within the status quo stock of knowledge is necessary to induce a willingness to learn. Closely related, but in a strictly neoclassical framework incorporating issues of rent-seeking, PRESCOTT (1998) and PARENTE and PRESCOTT (2000) argue that inefficient political institutions and mechanisms of corporate governance act as barriers that prevent the appropriation of knowledge spillovers in some economies.
- The relationship between the recipient and the source of the new knowledge is also of some relevance for the diffusion process. Only explicit knowledge can be communicated over great distances, while the transfer of tacit knowledge, if at all possible, involves direct interaction and therefore close spatial

proximity (ANSELIN *et al.*, 1997).⁸ One can therefore expect that explicit knowledge spreads much more rapidly over great distances compared with tacit knowledge.

The final point hints at the importance of locally confined innovative networks that are of special relevance for the diffusion of tacit knowledge. Whenever direct interaction is a necessary prerequisite for the diffusion of knowledge, it is obvious that space does indeed matter simply in the sense that the cost of setting-up and maintaining a direct interaction is likely to rise with the distance between the source and the recipient of the new knowledge. Intuitively, this can serve to explain why innovative networks often do not stretch across regional boundaries and why they are often relatively stable once established (WILKINSON and MOORE, 2000).⁹ The specific modes of local interaction between, for example, entrepreneurs, venture capitalists, universities and government agencies leads to specific informal institutions guiding innovative activity. Information about novelties flows more easily among agents located within the same area thanks to social bonds that foster reciprocal trust and frequent face-to-face contacts. Therefore, geographical clusters offer more innovation opportunities than scattered locations. Innovation diffusion is also faster (BRESCHI and LISSONI, 2001). Consequently, innovation processes are likely to differ across regions because different routines of interaction are established between the various parties involved in the process.

With these considerations, a gap between this class of approaches and the standard neoclassical literature becomes visible. The theoretical considerations generally do not involve optimizing individuals, but boundedly rational agents whose decisions are influenced by informal institutions, which are understood as patterns of thinking and decision-making that are shared among a larger group of individuals. Whether individual decisions yield efficient results then depends not only on technological knowledge (i.e. on the instalment of an efficient production technology), but also on the degree to which a certain set of institutions allows for the efficient use and extension of technological knowledge (for an economic historian's point of view on this problem, see, for instance, LANDES, 1998). In other words, the willingness and ability to absorb knowledge about new production technologies is quite likely to differ between regions depending on their actual institutional framework. While some innovative regions can be expected to exploit the information conveyed by knowledge spillovers eagerly, others may simply ignore it or lack the adequate institutional framework to transform the received knowledge into actual production technologies. Innovative networks are therefore characterized not only by stable routines to share knowledge internally, but also by routines of receiving and handling incoming knowledge spillovers.

Knowledge spillovers, agglomeration and regional economic growth

The above discussion has focused on micro-level conditions for and barriers to the spatial diffusion of knowledge. Reasons for an unequal spatial distribution of knowledge have been given, which imply that the range of knowledge spillovers may be limited and that regional differentials in the efficiency of using knowledge spillovers can be expected. If this is the case, then it is necessary also to have a more detailed look at the effects that knowledge spillovers in turn may have on both agglomeration and regional economic growth. Clearly, the mere existence of locally confined knowledge spillovers offers an incentive to locate production facilities such that those spillovers can be appropriated (GROSSMAN and HELPMAN, 1991). If, on the other hand, they were not locally confined, one would need to look elsewhere to explain agglomeration (KRUGMAN, 1991). Closely related influences that counteract incentives to cluster productive facilities, such as trade costs (FUJITA and THISSE, 2002), also need to be recognized.

On balance, however, the opportunity to appropriate knowledge spillovers often appears to outweigh such counteracting influences and gives way to some form of cooperation between firms. This cooperation can occur either informally, e.g. through regional clustering, or through formal cooperation in R&D activities. For example, SUZUMURA (1992) develops a model of firms who compete in an oligopoly on the product market and shows that while uncoordinated R&D spending with free-riding on sufficiently large spillovers will result in a level of spending that is lower than the welfare-maximizing level, a cooperative and enforceable commitment to R&D spending levels leads to overall spending that is generally higher than the non-cooperative solution but still socially insufficient. This reaffirms earlier results developed by D'ASPREMONT and JACQUEMIN (1988, 1990) in a duopoly model and is supported by empirical evidence for Germany (KAISER, 2002a). In addition to these results, KAMIEN and ZANG (2000) demonstrate that explicit cooperation is necessary to realize spillovers if the firms decide about not only the quantity of R&D expenditure, but also their qualitative R&D approach. Competing firms that do not cooperate will choose different approaches because by assumption of the model knowledge spills over effectively only between firms that use similar approaches, and in a non-cooperative world firms prefer to impede their own knowledge from spilling over to a competitor.¹⁰ CASSIMAN and VEUGELERS (2002) find in data on the Belgian manufacturing industry that firms indeed do have a propensity to protect knowledge spillovers to competitors through shared suppliers or customers by formally cooperating with their suppliers and customers.

Returning explicitly to the spatial dimension of knowledge spillovers and their economic effects, the idea that agglomeration economies may have positive effects on productivity can be traced back at least as far as the work of MARSHALL (1890/1966). An indication for the specific relevance of knowledge as a driving force behind agglomeration of productive activities is the empirical observation made by KAHNERT (1998) that innovative production facilities with a high knowledge intensity and frequency of direct communication tend to be centralized in the core of agglomeration, while standardized, routine-based production facilities are often (re-)settled in more peripheral regions. Micro-level knowledge spillovers that occur due to spatial proximity appear to be deliberately exploited by clustering the relevant facilities geographically.¹¹

The growth effects of agglomeration are therefore potentially twofold. On the one hand, it may be a general prerequisite to appropriate knowledge spillovers at all, if they are indeed locally confined. On the other hand, agglomeration leads to a concentration of innovative, research-intensive production facilities that are seen as important contributors to the growth of per-capita incomes. The concentration effect is therefore a foundation for sustained differences in the economic development of centres and peripheral regions (BALDWIN and MARTIN, 2003).

The gravitational force exerted by knowledge spillovers may be quite strong. For example, ENGLMANN and WALZ (1995) build a theoretical model in which they show that even if one factor entering the production of a high-technology good is immobile, the existence of knowledge spillovers in combination with a non-tradable intermediate good suffices to concentrate the production of the high-technology good in a core region, leaving to the peripheral regions the production of traditional goods for which knowledge spillovers are unimportant. The divergent growth paths of centres and agglomerations are further reinforced if human capital also tends to be concentrated in the agglomerations, while routinized tasks for unqualified labour are allocated to the peripheral regions. Given the absence of decreasing returns of learning in the aggregate production function of agglomerations, knowledge spillovers in agglomerations can therefore be interpreted as a source of sustained regional economic growth (GLAESER *et al.*, 1992; FUJITA and THISSE, 2002) and as an explanation for sustained differential growth rates between different regions with different patterns of agglomeration.

It is useful to distinguish between different types of knowledge spillovers because different mechanisms through which a positive knowledge externality may foster growth are conceivable. The principal distinction is commonly made between a location externality and an urbanization externality (CARLINO, 1987; HENDERSON *et al.*, 1995; HENDERSON, 1997;

FELDMAN and AUDRETSCH, 1996; AUDRETSCH, 1998; KEILBACH, 1998; PARTRIDGE and RICKMAN, 1999). An example for location externalities is the so-called MAR-spillovers. The term 'MAR-spillovers' (GLAESER *et al.*, 1992, p. 1127), which was coined after three pioneering contributions from MARSHALL (1890/1966), ARROW (1962) and ROMER (1986), denotes a spillover between researchers, entrepreneurs and businesses within an industry. An often-cited empirical example is the concentration of suppliers of semiconductors and related technologies in Silicon Valley, CA (AUDRETSCH and FELDMAN, 1994; CARLINO, 1995). MAR-spillovers lead to learning processes such as those described above, where knowledge spills over between individuals working to solve similar or at least related problems. MAR-spillovers are, therefore, intra-industrial phenomena that allow the exploitation of regional economies of scale: the relatively small technological distance (GRILICHES, 1979) between individuals and firms implies low barriers for knowledge spillovers and is seen as a condition for sustained growth. A testable hypothesis deduced from this approach would, therefore, state that regions characterized by a higher concentration of firms employing similar production technologies ought to, all things being equal, have higher income growth rates than regions with a lower concentration of similar firms.

Contrary to this class of spillovers, urbanization externalities denote the effect of the size and heterogeneity of an agglomeration. An example is the concept of the so-called Jacobian spillovers (JACOBS, 1970, 1986) that occur between different industries, leading to the exploitation of regional economies of scope. The mechanism in which this type of spillover works can be understood as a widening of the scope of research of individual industries through the interaction with other industries. An important difference between the two types of spillovers, as far as their incentive effects are concerned, is that heterogeneous businesses are often not in competition with each other and therefore may be more willing to engage in interactions less reluctantly than in the case of MAR-spillovers. As a testable hypothesis, it follows from research on Jacobs-spillovers that agglomerations with a high degree of diversity, all things being equal, ought to enjoy higher income growth rates than regions with a more homogeneous population of firms.

In both cases, it is important to note explicitly again that the absence of decreasing returns to scale that can result from the existence of knowledge spillovers is to be looked for in the aggregate production function of a region as a whole. Spillovers enter the single firm production functions, which can be conceptualized by letting the output of one firm depend on the economic activity of all other firms that operate within a defined spatial and/or technological distance. If they are quantitatively sufficient, these positive feedbacks between firms may then achieve the absence of decreasing

returns on the aggregate level, but leave the single firm facing decreasing returns.

EMPIRICAL STUDIES OF KNOWLEDGE SPILLOVERS

In the theoretical literature surveyed above, there are only few instances where the propositions of different approaches completely contradict each other. An example was discussed above. More often, though, the predictions derived from different models differ gradually or are deduced from differing assumptions whose empirical relevance relative to each other is disputed. In any case, it is necessary to complement the survey of the theoretical literature on knowledge spillovers with a survey of the results of empirical studies conducted on this issue so far. There are a considerable number of studies intended to deliver an empirical scrutiny of the theoretical literature on economic growth and taking knowledge into account. See, for instance, the seminal study by MANKIW *et al.* (1992), who show that a neoclassical growth model with diminishing returns, but accounting for human capital, does well in explaining cross-country differences in output per worker. However, it can be argued that the evidence produced by Mankiw *et al.* and subsequent attempts to reproduce this study are also consistent with endogenous growth theory (AGHION and HOWITT, 1998, ch. 12). Furthermore, COE and HELPMAN (1995) show for a dataset of 24 developed economies that a country's own total factor productivity rises with the stock R&D investments in other countries. This supports the presumption that knowledge spilling across regional or national boundaries does have an impact on growth.

With the jury still being out regarding the relative empirical performance of neoclassical and endogenous growth theories, it therefore appears to be a reasonable approach to investigate knowledge externalities in their own right.¹² To do so, it is not sufficient to demonstrate a geographical clustering of firms belonging to the same or to related industries. Such a clustering may not be the consequence of knowledge spillovers at all, but simply the result of natural advantages of certain areas. And, indeed, ELLISON and GLAESER (1999) demonstrate that about one-fifth of geographical clustering of industrial production in the USA can be explained by an (even incomplete) set of natural advantages. MORRISON PAUL and SIEGEL (1999) use a dynamic cost-function approach to find evidence for economies of scale in US manufacturing industries, and also show that external effects (the level of activity of other firms in the same sector entering a firm's production function) do well in explaining these economies of scale. The conjecture that knowledge externalities are the driving force here remains speculative, however, and cannot be directly inferred from their available data.

Therefore, to demonstrate the empirical relevance of knowledge spillovers, one has actually to show that knowledge externalities are being appropriated by firms in significant dimensions. On a general methodological level, two classes of approaches can be distinguished: one observing micro-level data, the other taking a focus on aggregate data.

The first class is characterized by the attempt to reconstruct actual paths of knowledge diffusion, e.g. by means of researching citations of patents and their spatial distribution. This allows empirical statements about the spatial range of knowledge spillovers that have actually occurred. This research strategy is essentially an attempt to follow the paper trail left by knowledge spillovers: citations in patents are added not by the inventor, but by specialized bureaucrats (patent examiners) to indicate which earlier ideas from patents have contributed to the newly patented invention. As, for example, the work of JAFFE *et al.* (1993) shows, the data on patent citations can also be exploited statistically. Jaffe *et al.* calculate two probabilities: that a patent cites another patent registered by a nearby university or firm, with both patents referring to the same technology and having originated from a similar time; and that two patents are similarly geographically linked, without there being a formal link through a patent citation. The second probability is supposed to control for other rationales of regional clustering. If the first probability is significantly higher than the second, this is interpreted as evidence supporting the hypothesis that knowledge spillovers are indeed being appropriated by regionally proximate firms or research institutions. If it can also be shown that the probability of citation decreases with distance, then this is supporting evidence for a limited spatial range of spillovers. A different micro-level approach are surveys of firms questioned about the sources of their stock of knowledge, or, more specifically, about the impact of research conducted outside the firms, e.g. in universities or in other firms, on their own production processes. The data generated by such surveys then allow direct inferences about the relative importance of knowledge spillovers in different sectors, and about the impact of distance on the ability to appropriate spillovers.

As an example for the second class of studies, there is a class of approaches that aim to estimate a knowledge production function. A pioneering study in this respect is by GRILICHES (1979). In these approaches, a dependent variable such as the number of corporate patents in an industrial sector in a particular region is regressed on a number of independent variables that may include research expenditures by other industries in the same region, research expenditures by academic institutions in the region and, of course, a number of control variables. Therefore, these approaches generally allow testing of hypotheses about the impact of spillovers from academic research and also about the existence of Jacobian spillovers, i.e. the hypothesis that

research expenditures from other sectors increase innovative activity. Again, by also taking research expenditures in geographically more distant regions into account, statements about the range of knowledge externalities are possible.

Closely related, the search for spatial autocorrelation has emerged as an approach to the exploratory analysis of regional interdependence. For example, a correlation coefficient proposed by MORAN (1959) that has been introduced into spatial economic analyses takes the form:

$$c = \frac{N}{N_n} \frac{\sum_{i=1}^N \sum_{j=1}^N (x_i - \bar{x})(x_j - \bar{x})w_{ij}}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

where N is the number of regions in the sample and N_n is the number of neighbouring regions, x is the economic variable under consideration in regions i and j , and w_{ij} is a spatial weight that takes a value according to the relationship between regions i and j . For example, in a simple binary analysis, the weight takes the value of 1 if regions share a border and zero if they do not. In a more sophisticated analysis that also aims at making inferences about the range of spillovers, the weight may also be calculated such that its value contains information about the distance between two regions. If the economic variable under consideration is, for example, the number of patents granted to firms in a particular sector, then a significantly positive value of the correlation coefficient would support the hypothesis that knowledge spillovers with limited spatial range exist between these regions.

In further applications, a fruitful approach has been to incorporate spatial autocorrelation into studies of regional convergence.¹³ An obvious objection against these approaches could be the argument that it is prone to misspecifications in the sense that the regional boundaries (e.g. administrative boundaries) that enter the empirical analysis are not identical with economic regions, which may, for example, overlap state or county borders. However, it is possible to identify such nuisance dependence and account for it (ANSELIN and REY, 1997).

The following subsections present the main empirical results about the existence, range and economic impact of knowledge spillovers found in the literature.

Impact of distance

The heterogeneity of empirical approaches is mirrored to a certain degree in the heterogeneity of results produced by these approaches. Despite the methodological problems and differences, there appears to be a widespread consensus that spatially confined knowledge-spillovers are an important empirical phenomenon with a significant impact on economic performance (e.g. JAFFE *et al.*, 1993; AUDRETSCH and FELDMAN,

1994; FELDMAN and AUDRETSCH, 1996; HENDERSON, 1997; FRITSCH and LUKAS, 1998; KEILBACH, 1998; FUNKE and NIEBUHR, 2000; NIEBUHR, 2000; PACI and PIGLIARU, 2001). However, no consensus is reached about the spatial range that can be attributed to knowledge spillovers, and in fact the majority of studies refuse to quantify the range at all.

Notable exceptions are ANSELIN *et al.* (1997), VARGA (2000), KELLER (2002) and BOTTAZZI and PERI (2003). Anselin *et al.* analyse the impact of university research and private R&D activities on regional innovation patterns for data from high-technology firms in the USA. This is done with cross-section data for states and metropolitan statistical areas in the country, which are used to estimate a knowledge production function where industry and university R&D are the independent variables. The authors find a significantly positive impact of university research on innovative activity within a range of 50 miles around the university that is the source of new knowledge. For private R&D activities, however, no significantly positive result could be produced. This supports the theoretical considerations about the different incentives within the public and private sectors to circulate newly generated knowledge. In addition to these results, Varga, who also uses a knowledge production function framework, shows that knowledge spillovers do not only occur within metropolitan areas. Rather, spillovers into neighbouring metropolitan areas up to 75 miles away also have a significant positive impact on innovative activities in these neighbouring areas.

Bottazzi and Peri focus on data for European regions over 18 years to 1995. Choosing the number of patents per square kilometre of a region as the dependent variable, they estimate the influence of a region's own R&D spending (per square kilometre) and the influence of R&D spending within distances of 0–300, >300–600, >600–900, 900–1300 and >1300–2000 km away from that region. For knowledge spillovers resulting from R&D investments and patent applications, a significant positive impact on innovative activities in neighbouring regions appears to exist only for a distance of up to 300 km. However, the magnitude of this impact is relatively small. An increase of R&D spending of 100% in one region can be expected to yield an increase of innovative activities of 80–90% in the home region, but only of 2–3% in a neighbouring region. Keller takes an international rather than a regional economic perspective. Using data on manufacturing industries in 14 Organization for Economic Co-operation and Development (OECD) countries, he estimates the impact of one's own R&D spending and that of the G5 countries (France, Germany, the UK, Spain and Italy) on total factor productivity in the 14 OECD countries. The results support the hypothesis that knowledge spillovers are localized, with the impact of foreign R&D spending on productivity being reduced by one-half at every 1200 km

at best. However, Keller also shows that knowledge spillovers have become less localized over time.

That most other empirical studies do not quantify a spatial range of knowledge spillovers simply follows from the decision of their authors to define their units of research in such a way that the distance of spillovers is implied by this definition. For example, JAFFE (1989) and FELDMAN (1994) are interested in spillovers between US states; and PERI (2003) researches the diffusion of knowledge between predefined regions in Europe and the USA. What is of interest then is the effect of a spillover from a predefined region into a neighbouring region, but a quantifiable distance between neighbouring regions simply does not exist. Nevertheless, the limited spatial range of knowledge spillovers can also be shown in studies of this type, as AUDRETSCH and MAHMOOD (1994) show. In addition to spending on university research and private R&D, Audretsch and Mahmood also use the number of research institutions in a city and in a state as independent variables. While the number of research institutions in a city has a significantly positive impact on innovations in that city, the number of institutions in the same state does not. The authors argue that this implies that knowledge spillovers are a local phenomenon.¹⁴

Empirical results supporting this view have also been obtained for German data. FUNKE and NIEBUHR (2000) and NIEBUHR (2000) show in a cross-sectional analysis for 75 regions that knowledge spillovers predominantly occur between neighbouring regions. They find that for every 23–30 km from the source of a knowledge spillover, the positive effect of this spillover on innovative activities decreases by 50%. In addition, they also find supporting evidence for the theoretical prediction that innovative activities are usually concentrated in agglomerations and are less relevant for peripheral regions. An increase of productivity because of a knowledge spillover from such an innovating agglomeration is restricted to regions in the immediate neighbourhood. Similar results are produced by BADINGER and TONDL (2002) in a study on the determinants of regional growth in 159 regions within the European Union. According to them, an inflow of knowledge has a positive impact on the growth of a region, but this effect has a larger magnitude if neighbouring regions exhibit high growth rates themselves. In this context, studies by GREIF (1998) and the FRAUNHOFER-INSTITUT FÜR SYSTEMTECHNIK UND INNOVATIONSFORSCHUNG (2000) are also interesting, although they do not focus on knowledge spillovers in the narrower sense. However, they do show that indicators for the spatial distribution of knowledge generation such as the number of R&D employees and patent applications exhibit a high degree of regional concentration.¹⁵ This finding contradicts with the assumption of an instantaneous and global diffusion of knowledge, because in this case similar innovative

reactions to knowledge spillovers would have to occur in all regions.

Sectoral differences and firm size

From an even more disaggregated perspective, knowledge spillovers appear to be particularly relevant in 'young' industries and sectors where new knowledge can be assumed to be of special importance. This effect has been shown to exist by, among others, GLAESER *et al.* (1992) and FELDMAN and AUDRETSCH (1996). The intuitively plausible result that sectors with a heavy reliance on knowledge are also more sensitive to knowledge spillovers has been obtained by AUDRETSCH and FELDMAN (1994). The life cycle effect is also in line with intuitive reasoning, since it is easily conceivable that relatively young sectors with a low fraction of routinized activities exhibit a greater demand for new knowledge and a greater willingness to use new incoming knowledge compared with mature sectors where a large fraction of activities is already following established routines that are costly to change. Furthermore, firms in the first periods of their life cycle often do not have the capacities to maintain large-scale R&D departments themselves and, therefore, rely on external sources of knowledge to a larger extent than more mature firms with extensive own R&D activities.

Similar mechanisms are at work for smaller-scale firms that often do not have sufficient own resources for extensive R&D activities. LINK and REES (1990) survey 158 firms involved in cooperation programmes with universities and show that such programmes have a significantly positive effect on the innovative activities of small firms. The surprising result of this survey is that the fraction of large firms taking part in such programmes is larger than the fraction of small firms. The smaller firms taking part have a lower ratio of R&D activities to total revenue, however. Apparently, smaller companies seem to make use of the programmes more efficiently, substituting their own R&D by incoming knowledge spillovers. It is therefore somewhat paradoxical that large firms enter such programmes with a greater probability than small firms.

The negative effect of knowledge spillovers on the cost of innovating is also confirmed in other studies (e.g. BERNSTEIN and NADIRI, 1988, 1989; LEVIN and REISS, 1988). FELDMAN (1994) inquires into the influence of technological infrastructure on private-sector innovations. As indicators for technological infrastructure, private sector and university R&D outlays, the embeddedness into up- and downstream production and the existence of firm-related services have been used. Quite in line with the other studies mentioned here, Feldman concludes that primarily small firms make use of external resources as additional inputs in order to compensate for a lack of their own resources compared with larger firms. Other empirical studies with similar results are by ACS and AUDRETSCH

(1988), AUDRETSCH and ACS (1991) and AUDRETSCH and MAHMOOD (1994). FRANKE (2002) reproduces these results for three German regions, relying on a survey of 1800 firms. Franke also shows that both intra- and inter-industrial knowledge spillovers have a positive impact on the number of patents applied for by small- and medium-sized companies. The same holds for knowledge spillovers out of universities.

Importance of different types of spillovers

The evidence concerning the relative importance of different types of knowledge spillovers is somewhat more ambiguous. However, there are empirical studies that support the hypothesis that Jacobian spillovers do matter, as well as those that show the same for MAR-spillovers. Therefore, even if the relative importance of both appears to be difficult to sort out, the statement that both types of knowledge spillovers are empirically relevant is a rather safe claim (HENDERSON, 1997; PARTRIDGE and RICKMAN, 1999; FORNI and PABA, 2001). One of the few studies accounting for both types of spillovers is by KELLY and HAGEMAN (1996), who attempt to show for US state-level data that there is a positive impact of both inter- and intra-industrial spillovers on the number of patent applications (FELDMAN and AUDRETSCH, 1999).¹⁶ Kelly and Hageman show that a local transfer of knowledge has a positive effect for 11 of 12 industrial sectors under observation, while intra-industrial knowledge spillovers could be shown to have a significantly positive effect only in two of 12 sectors. This result casts some doubt on the relative importance of MAR-spillovers, which predict that knowledge spillovers will occur within a group of relatively homogeneous firms.

Some doubt on the relative dominance of Jacobian spillovers is, however, cast by BERNSTEIN and NADIRI (1988), who look at knowledge spillovers between US firms producing in different sectors between 1958 and 1981. It was shown that there are significant differences in the spreading of one's own knowledge as well as in the form of reception of new, externally produced knowledge between firms working in different sectors. An important result of their study is that firms in four of five surveyed sectors are sources of knowledge spillovers, but only in two sectors do firms act as recipients of such spillovers.

This finding hints at the existing difficulties of generalizing results about the relative importance of both types of knowledge spillovers. Which kind of knowledge spillover is dominant (or used at all) depends eventually on micro-level, i.e. sectoral and firm-level, conditions (PORTER, 1990), which are not sufficiently researched thus far. There are few hints at possible criteria that allow a prediction of the relative importance of both types. One is the presumption that firms that tend to conduct their R&D activities in a more incremental way have a propensity to rely more on

intra-industrial knowledge spillovers, while firms working on more groundbreaking innovations exhibit a larger dependence on inter-industrial externalities. Similarly, one might expect that the intensity of competition within a sector and the degree of vertical integration of production have adverse effects on the frequency of intra-industrial knowledge spillovers.

Transmission channels and absorptive capacity

The evidence regarding the relative ease with which knowledge spillovers diffuse across organizational boundaries is, fortunately, less ambiguous. Recent studies on this issue, which lay some emphasis on the receiver's capacity to appropriate spillovers, are by MARIANI (2000), VERSPAGEN and SCHOENEMAKERS (2000) and MAURSETH and VERSPAGEN (2002).¹⁷ An important result of these studies is that knowledge is absorbed relatively easier in regions that already have a relatively higher productivity level and a larger stock of knowledge. Furthermore, and quite in line with intuition, technological proximity (i.e. a recipient of a spillover commanding expertise in the same or related fields as the sender) also has a positive impact on the amount of knowledge that spills over from one organization to another. This is clearly supporting evidence for the conjecture that knowledge is acquired in a cumulative process during which new incoming knowledge can only be used if necessary complementary knowledge is already in use. The diffusion between regions occurs easier if the source region and the receiving region resemble each other in terms of their sectoral structure. This again supports the conjecture that the appropriation of a knowledge externality is more likely to be successful if the source and the recipient are characterized by similar histories of knowledge accumulation. However, it also supports the presumption that between regions MAR-spillovers spread easier than Jacobs-spillovers, since the heterogeneity of regions cannot be shown to facilitate a frictionless diffusion of knowledge over regional boundaries in empirical studies.

A look at the transmission channels shows that knowledge spillovers occur particularly often within multinational companies, which is not surprising since adverse influences on diffusion such as a competitive relationship or varying routines underlying R&D activities generally do not play a role if knowledge diffuses within the same company. There is some evidence that firms deliberately locate research activities such that they can use knowledge and skills found in specific regions. For example, CANTWELL and JANNE (1999) argue empirically that leading multinational companies tend to conduct parallel but complementary innovative activities in foreign-located facilities. They also show that there is a flow of knowledge in the other direction, with firms from regions where very specialized knowledge is available locating some of their research in

higher-order centres in order to use the more general expertise available there. A significantly positive effect of foreign direct investment on technological progress in the host country has been found by BARRELL and PAIN (1997). In a similar vein, DRIFFIELD and MUNDAY (2000) find that the comparative advantage of UK manufacturing industries significantly increases with inflowing foreign direct investment. However, LUTZ (2000), TYBOUT (2000) and XU (2000) argue that knowledge diffusion within multinational companies is often confined to countries with relatively high income levels. An intuitive explanation for this fact might be that R&D activities relying on highly qualified labour are usually settled in highly developed economies.

Beyond this special case of intra-firm knowledge transmission, inter-firm knowledge spillover or a spillover between firms and public sector research are other important transmission mechanisms. As the literature reviewed by FELDMAN (1999) shows, these spillovers often occur through direct interaction on the individual level. SCHRADER (1991) also shows that the frequency of interaction of R&D employees from different firms has a positive impact on the frequency of innovations in these firms. To set up these interactions, it was shown to be of pivotal importance that the individuals expected the relationship to be reciprocal regarding the quality and quantity of knowledge that was to be exchanged. Whenever such a reciprocal relationship was not expected by an individual, he/she refused to act as a source of knowledge spillover. Thus, this evidence can also be interpreted cautiously as supporting the hypothesis that the perceived stability and reliability of given social networks between individuals support the diffusion of knowledge.

More direct evidence for the importance of social networks for knowledge spillovers and the frequency of innovations can be found in a pioneering study by PIORE and SABEL (1984), who analyse the determinants of innovative activities in industrial districts in north-eastern Italy and find supporting evidence for the hypothesis that stable networks between firms lead both to an increase of innovative activity and to a reduction of transaction costs. The concept of industrial districts has some history in economics. It was coined by MARSHALL (1890/1966) to denote an agglomeration of specialized industries at particular locations, where the agglomerated industries are primarily small- and medium-sized firms, often family businesses. The firms in these industrial districts are usually active in sectors characterized by low or no economies of scale at the firm level, which implies that a market structure with many relatively small suppliers can be efficient.

The results of Piore and Sabel have been reproduced by a large number of studies concerning industrial districts in Italy (e.g. LAZERSON, 1993, 1995; GOTTARDI, 1996). Despite controlling for a large number of other influences, local and regional networks have been

shown to exert an essential, significantly positive impact on innovative activity in all surveyed districts. In the meantime, other studies have shown this is not solely an Italian phenomenon: SAXENIAN (1994) produces evidence supporting the influence of networks in Silicon Valley and Boston, MA; GARNSEY and CONNOR-BROOKES (1993) find the same for Cambridge, UK. GANNE (1992), KRISTENSEN (1992), MASKELL (1992) and SAGLIO (1992) find similar evidence for both Denmark and France. Given the structural dissimilarities of all these countries and regions, a general importance of networks appears to be supported by the evidence.

ZUCKER *et al.* (1998) focus on the relevance of human capital for knowledge spillovers in a study of the geographical location of biotechnology firms. Their main interest is on the very early stages of innovations in this sector, where results of scientific research in universities or research institutions are transformed into commercial products. Behind their empirical study lies the assumption that specialized knowledge is embodied in highly qualified individuals. They show that the geographical location of firms is closely correlated with that of star researchers in the field. However, it is also shown that the external knowledge relinquished by a cooperating scientist is used exclusively by the company with which this scientist was involved. In other words, there are no knowledge spillovers between firms regarding this external knowledge. The authors explain this fact by arguing that the knowledge surrendered by the scientists is implicit, embodied knowledge that is directly bound to the person. They find that these technical properties of the new knowledge used by the companies make further knowledge spillovers unlikely or even impossible. In the same vein, ALMEIDA and KOGUT (1999) focus on the mobility patterns of individual patent-holders (engineers) in a number of industry clusters. They find that mobility is high and highly localized, but only in Silicon Valley, which is also the only cluster wherein such mobility positively affects the innovation rate of local firms. The particularly high mobility of skilled labour between firms in Silicon Valley is attributed to 'social institutions' that are particular to that region, without giving an explicit account of what exactly are these institutions.¹⁸

As the discussion in the second section has shown, knowledge is not necessarily embodied in persons, but can also be embodied in products. LEVIN *et al.* (1987) argue, based upon US data, that reverse engineering of new products from competing companies is an important source of knowledge spillovers. HARABI (1997) comes to similar conclusions by surveying 358 Swiss companies for their strategies of gaining new technological knowledge. Both studies have shown that beyond reverse engineering, frequently used strategies to acquire knowledge are the licensing of technologies, surveys of the (scientific) literature and

communication with employees of competing businesses. For Italian data, NAPOLITANO (1991) finds similar results. The open recruitment of employees from competing companies and research in patent databases, in contrast, play only a minor role.

A number of studies are interested in the importance of public-sector research institutions as a mechanism for regional knowledge spillovers, with JAFFE (1989) being a somewhat representative study. Jaffe analyses knowledge spillovers by using a 'Geographic Coincidence Index', which is intended to measure spatial clustering of private and university R&D spending. With data from US 29 states, Jaffe analyses such geographical clustering and finds support for the conjecture that university research and private R&D are closely correlated in the spatial dimension. Several further empirical studies (e.g. LINK and REES, 1990; AUDRETSCH and MAHMOOD, 1994; AUDRETSCH and STEPHAN, 1996; ACS *et al.*, 1999) confirm this result for the USA. For German data, NERLINGER (1996), FRITSCH and SCHWIRTEN (1998), STERNBERG (1998), EDLER and SCHMOCH (2001) and FRANKE (2002) find a similar correlation: the more a region is endowed with university research or other public-sector research institutes, the more clustered are private R&D activities in this region, and the more likely is the establishment of new R&D-intensive firms.

Why networks matter

The above discussion has already hinted at the fact that certain configurations of firms and research institutions or of cooperating small businesses may advance the use of knowledge as a shared resource. One recurring concept in the literature on such phenomena is that of (social) networks. For example, it has been shown that venture capitalists tend to invest primarily in firms located close to their own offices (SORENSEN and STUART, 2001) and that they have good reason to do so because geographically proximate investments generate abnormally high profits (COVAL and MOSKOVITZ, 2001). Closely related, STUART and SORENSEN (2003) show that new biotechnology firms in the USA are more likely to be founded close to venture capitalists, universities and incumbent firms. Even if the direction of causality is unclear – do firms follow venture capitalists or do venture capitalists set up offices where they expect research institutions to attract start-up businesses? – it appears to be evident that geographical proximity does matter for these firms.

ZUCKER *et al.* (1998) show that locations where star scientists of their field reside are particularly attractive for biotechnology firms to take residence. Moreover, firms that have established formal relationships with these scientists improve their growth rates significantly. Therefore, it appears that when formally organized, intentional knowledge spillovers are a driving force in the establishment of networks of firms. The relevance

of this fact not only has been shown for modern technologies, but also has been demonstrated by LAMOREAUX and SOKOLOFF (1999) for US glass manufacturers in the pre-First World War era. SORENSON (2005) looks at the entrepreneurial process to explain the emergence of firm networks and argues that personal social networks among managers of already existing firms allow an exchange of information on the relative profitability of their activities. Informal, face-to-face communication between individuals who are already mutually acquainted allows the identification of entrepreneurial opportunities. When a new business is founded, choosing a location where similar firms already exist allows access to a local labour market with individuals who have already acquired relevant tacit knowledge, or knowledge that is embodied in individuals who are themselves relatively immobile geographically (ALMEIDA and KOGUT, 1997).

Networks of regionally clustered businesses and institutions, therefore, offer two broad opportunities: formal exchanges of knowledge through market relationships, where proximity allows the establishment of closer ties; and the informal exchange of knowledge in social networks of individuals. These considerations are consistent with empirical findings that, at a city level, cities that are better endowed with human capital have higher sustained growth rates (GLAESER *et al.*, 1992). Cities or regions with skilled labour and high levels of specialized human capital are more likely to attract innovative networks compared with less well-endowed areas, simply because firms intentionally exploit knowledge flows that are regionally bounded.

CONCLUSIONS AND OPEN QUESTIONS

The literature surveyed in this review offers several insights into the properties of knowledge that may be surprising to a neo-classical economist who has not been familiar with this literature before. The general insight is that knowledge does not diffuse frictionlessly. Even for knowledge that *prima facie* appears to be perfectly non-excludable, such as the published results of academic research, the spatial dimension has been shown to be important: Start-up firms in sectors such as biotechnology tend to settle in close proximity to renowned researchers in this area and seek to establish formal relations with these individuals in order to appropriate knowledge spillovers. Similarly, the discussed studies that attempt to follow the paper trail left behind by knowledge spillovers (in the form of patent citations) show that there is a significantly positive relationship between geographical proximity and the probability of a patent citation.

The empirical evidence, therefore, strongly supports the theoretical assumption that knowledge used in economic processes exhibits a certain degree of tacitness. Even perfectly non-excludable knowledge from

scientific research may be easier (less costly) to appropriate if the individual who has produced it is personally available to give advice on how to apply it in actual productive processes. This tacitness gives firms an incentive to form clusters, and the emergence of clusters has immediate implications for the differential development of different geographical areas. With a concentration of knowledge-intensive, high-technology production facilities in core regions on the one hand and peripheral regions with production processes employing relatively abundant unqualified labour on the other hand, sustained differentials in income growth are likely. In fact, the two types of places can even be thought of being subject to different growth regimes: knowledge-driven endogenous growth without diminishing returns in the aggregate production function and a standard neoclassical decreasing-returns regime.

Given these considerations, it is not surprising that firms deliberately attempt to exploit spatially limited knowledge spillovers. However, they face a dilemma if they do so. While they obviously would like to appropriate relevant knowledge spilling over from other firms, they have an incentive to protect their own stock of knowledge against competitors. The theoretical literature suggests that they can do so by negotiating optimal contracts for their employees that give them an incentive to remain with the firm as long as this is efficient. The empirical literature, on the other hand, suggests that informal communication between employees that is difficult to observe by their firms plays a role. In this sense, optimal contracts are probably not perfectly enforceable. The literature on social networks suggests that such informal communication is even a mechanism for individuals to identify profit opportunities that are not yet used and initiate own start-ups to exploit these opportunities. Geographical proximity and the possibility for informal communication is then an important factor for generating market dynamics.

But such dynamics do not only result from knowledge spillovers between competing firms. As the literature on industrial districts shows, cooperation between relatively small geographically proximate firms can also be important. Small firms whose production functions are not characterized by economies of scale and who do not compete in the same market can increase the innovativeness by cooperating with each other. Again, the increased rate of innovation can on average be expected to generate faster income growth in such industrial districts, compared with places where this form of cooperation does not exist.

Despite the large number of surveyed contributions, both theoretical and empirical, there still appear to be a number of unanswered questions. The following points are of particular importance in our view:

- The empirical studies so far do not offer clear guidance about the question of 'the role of geographical

distance in the economics of knowledge transmission, which is still rather controversial' (BRESCHI and LISSIONI, 2001, p. 976). All the best-known studies on knowledge spillovers seem to be unanimous in concluding that such spillovers are important and bounded in space. However, the evidence on geographical knowledge spillovers is largely of an indirect kind and cannot be taken as definite. Following BRESCHI and LISSIONI (2001), the major limitation of the empirical literature is that virtually no contribution has explored the ways in which knowledge is actually transferred among people located in the same geographical area. There is hardly any doubt that innovation networks are often locally confined, but the rationale for this may have less to do with the limited spatial range of knowledge spillovers and more with, for example, the need to establish transaction-intensive relationships with suppliers and customers. Against this background, one could argue that the concept of geographical knowledge spillovers is still no more than a 'black box'.

- Another question regards the distinction between tacit and codified knowledge that plays a central role in much of the literature on knowledge spillovers. Following COWAN *et al.* (2000), technical knowledge (and even more scientific knowledge) may be considered as 'tacit' not because it cannot be articulated (as, for example, the often-cited craftsman's knowledge), but because it is highly specific in the sense that the understanding of this knowledge (or the message that transports that knowledge) depends on the capability of the addressees to decode the messages, which varies according to their expertise in the field. When tacitness is classified not only as an intrinsic property of knowledge, but also as a property of the communication of messages itself within a specific (epistemic) community, one is bound to recognize that the enabling conditions for benefiting from knowledge spillovers are far more complex than that of physical proximity.
- Some further research appears to be necessary about the mechanisms and sources of regional knowledge spillovers. In the literature, the observed correlation between knowledge (indicated, for example, by R&D output) and innovation has frequently been assumed to reflect pure technology externalities. However, there may be other economic forces at work 'that are different in nature but observationally equivalent in the knowledge production function setting' (BODE, 2004, p. 45). Other economic forces may be pecuniary externalities resulting from labour pooling, or from knowledge flows mediated by market mechanisms or by clubs. Following BRESCHI and LISSIONI (2001), there is still a need to explore the price and non-price mechanisms through which knowledge may be traded between universities and firms, as well as between firms. Additionally, the influence of particular elements of

institutional conditions, such as different access to markets for factors and goods, public funding of private research, and active interventionist policies on knowledge spillovers is not sufficiently researched thus far, and only few first steps have been taken (e.g. FELD *et al.*, 2003).

- Finally, the normative question is not settled whether the theoretical considerations and empirical evidence produced so far warrant any economic policy measures actively encouraging knowledge spillovers and, if so, how such policies ought to be designed. Some studies display an uneasiness with pure laissez-faire policies (e.g. MATSUYAMA and TAKAHASHI, 1993), but there are no well-funded prescriptions for active interventions either. Depending on the specific economic forces mediating knowledge flows, the policy conclusions may differ considerably. Against this background, the current (and still increasing) popularity of the concept of geographical knowledge spillovers may 'lead to naïve policy implications, which recall many not-so-distant and unfortunate experiences with science parks, growth poles and the likes' (BRESCHI and LISSIONI, 2001, p. 977).

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NOTES

1. For contributions emphasizing the relevance of knowledge from both a micro- and macroeconomic perspective, see, for example, NELSON (1982), AUDRETSCH and FELDMAN (1986), JAFFE *et al.* (1993), CARLINO (1995), MATUSIK and HILL (1998), CANIELS (2000), DOHSE (2001) and METCALFE (2002).
2. For early contributions to the endogenous theory of growth, see LUCAS (1988) and ROMER (1986); for a comprehensive survey, see FAGERBERG (1996). For some representative contributions to the literature on endogenous technological progress, see TALLMAN and WANG (1994), JAFFE (1998) and CANIELS (2000).
3. The importance of knowledge spillovers in these respects has been emphasized by, among others, BARRO (1991), GROSSMAN and HELPMAN (1991) GLAESER *et al.* (1992), GRILICHES (1992), HENDERSON *et al.* (1995), ANSELIN *et al.* (1997), FRITSCH and FRANKE (2000), KEILBACH (2000) and SMOLNY (2000).
4. As JONES (2002) argues, the path of the accumulation of knowledge is, however, never completely determined by the past, but is subject to random shocks.
5. For a distinction of different types of knowledge, see also MATUSIK and HILL (1998) and DOSI (1988).
6. For a general treatment of externalities, see SCITOVSKY (1952) and BAUMOL and OATES (1979).
7. For a model of leapfrogging, see BREZIS *et al.* (1993). The New Economic Geography is introduced and

- surveyed in KRUGMAN (1998a, b), SCHMUTZLER (1999), FUJITA *et al.* (2000), OTTAVIANO and THISSE (2003) and BALDWIN and MARTIN (2003).
8. For the importance of direct (e.g. face-to-face) communication, see also AUDRETSCH and FELDMAN (1996), AUDRETSCH and STEPHAN (1996), DOHSE (1996, 2001), FELDMAN and AUDRETSCH (1996), HENDERSON (1997), AUDRETSCH (1998), ANTONELLI (2000), CANIELS (2000) and CAPPELLIN (2001).
 9. According to CAMAGNI (1991, p. 74), innovative networks can be defined as 'the set, or the complex network of mainly informal social relationships on a limited geographical area, often determining a specific external "image" and a specific internal "representation" and sense of belonging, which enhance the local innovative capability through synergetic and collective learning processes'. See also BRÖCKER *et al.* (2003), in which several authors provide a timely and comprehensive picture on location, networks, and clusters as an important means in theoretical understanding of regional innovation and the factors influencing regional productivity and regional competitive performance. For a current examination of the relationship between network architecture and knowledge diffusion performance, see COWAN and JONARD (2004), who model knowledge diffusion as a barter process in which agents exchange different types of knowledge. For the relationship between network density and R&D spillovers, see MEAGHER and ROGERS (2004). Using ideas from organizational theory, they model how the structure and function of a network of firms affects their aggregate innovativeness. Two main results emerge: (1) the marginal effect on innovativeness of spillover intensity is non-monotonic; and (2) network density can affect innovativeness, but only when there are heterogeneous firms. The latter finding points out the relevance of so-called Jacobs-spillovers, which are discussed more intensively in the following subsection.
 10. KAISER (2002b) generalizes a model in the spirit of KAMIEN and ZANG (2000) by also taking into consideration that the degree of competition in the market for finished products supplied by the firms who may engage in R&D cooperation may indeed influence their willingness to cooperate.
 11. Obviously, there are also other economic arguments in favour of agglomeration, such as natural locational advantages, access to local markets, etc. (GLAESER *et al.*, 1992; JAFFE *et al.*, 1993; ANTONELLI, 2000; CANIELS, 2000; KEILBACH, 2000; CAPPELLIN, 2001).
 12. For a survey of measuring knowledge spillovers per se, without explicit reference to the regional aspects of the problem, see GRILICHES (1992). For recent accounts of the empirical literature of the geography of spillovers, see AUDRETSCH and FELDMAN (2004) and ROSENTHAL and STRANGE (2004), in both of which the literature on knowledge spillovers and the link between knowledge and agglomeration economies are surveyed from a more current point of view. For a critical re-examination of the empirical literature on localized knowledge spillovers and spatially limited innovation systems, see BRESCHI and LISSONI (2001). For a useful background discussion on the empirical measurement of knowledge spillovers, see KAISER (2002c).
 13. For more information on the empirical tools and methods regarding the econometric analysis of regional growth and convergence processes, see, for example, ANSELIN and BERA (1996), ANSELIN (1998), REY and MONTIURI (1999), FINGLETON (2000, 2001) and KOSFELD and LAURIDSEN (2004).
 14. As another example for this class of studies, see PERI (2003). Starting with the hypotheses that knowledge flows within and across countries should be carriers of important learning spillovers, Peri used data on 1.5 million patents and 4.5 million citations to analyse knowledge flows across 147 subnational regions. Peri found that only 15% of average knowledge is learned outside the average region of origin, and only 9% outside the country of origin. However, he also showed that knowledge in some sectors (such as computers) flows substantially farther, which is also the case for knowledge generated by technological leaders (top regional innovators).
 15. See also BODE (2004), who analysed the relevance of interregional knowledge spillovers as one determinant of innovative activity in West German planning regions in the 1990s. Bode found out that, in general, interregional spillovers can contribute significantly to regional knowledge production. Interestingly, the effects of knowledge spillovers are found to depend not only on the conditions prevailing in sending regions, but also on the conditions in recipient regions. The empirical results show that only a small fraction of the knowledge available in neighbouring regions actually spills over, whereas only regions with low R&D density benefit from interregional spillovers. For regions with high R&D density, they seem to be negligible. With focus on this type of regions, the overwhelming majority of innovative activity and resultant patents is produced by exploiting the region's own resources.
 16. To examine the role of specialization versus diversity, i.e. MAR- versus Jacobian spillovers, the authors tested whether the number of innovations from an industrial sector of a certain state owed more to the city specialization in this sector or to the precedence, within the state, of other industries whose science base was related to that of the examined industrial sector. A conclusion was reached that diversity matters more than specialization and, above all, this was interpreted as evidence that knowledge spills over across sectors rather than within them.
 17. These studies are based on counting the number of patent citations between pairs of regions and then estimating a (econometric) model where these counts are related to the geographical distance between pairs of regions. Their estimates show that the number of cross-citations drops significantly as the distance increases. In a similar manner, BROUWER *et al.* (1999) found that firms located in agglomerated Dutch regions tended to produce a higher number of new products than firms located in more peripheral regions.
 18. See also ZELLNER (2003), who presents evidence for embodied knowledge transfer via scientist migration. With empirical data on German scientists formerly employed by the Max Planck Society (MPS), Zellner tested the hypothesis that a substantial proportion of

the wider economic benefits to society from publicly funded basic research is associated with scientists' migration into the commercial sector of the innovation system. For evidence for embodied knowledge transfer,

see PARK (2004). Based on a data set of 21 OECD economies and Israel between 1971 and 1990, the study empirically explores the significance of student flows as a channel of R&D spillovers.

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